SULIT

CP115
Computer Science
Semester I
Session 2026/2027
2 hours 30 minutes

CP115
Sains Komputer
Semester I
Sesi 2026/2027
2 jam 30 minit

Nama	No. Matrik											Praktikum

(Sila isi maklumat dengan jelas dan lengkap)

(SET B)

KOLEJ MATRIKULASI JOHOR
JOHOR MATRICULATION COLLEGE

PRA-PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME TRIAL EXAMINATION

SAINS KOMPUTER

2 JAM 30 MINIT

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Untuk Kegunaan Pemeriksa		
No. Soalan	Markah	
	Pemeriksa	KP / KKP
1 (17 M)		
2 (10 M)		
3 (19 M)		
4 (10 M)		
5 (14 M)		
6 (10 M)		
JUMLAH (80 M)		

Kertas soalan ini mengandungi 10 halaman bercetak
This question paper consists of 10 printed pages

1. Problem solving in software development requires a meticulous process to ensure a reliable and efficient solution.

   a. State the first two (2) phases in the problem-solving cycle.

[2 Marks]

_____

_____

_____

_____

   b. Based on the situation related to software development below, state the correct step in problem solving for each statement.

Scenario	Step in Problem Solving
(i) The programmer interviews the client to determine that the system must be able to calculate monthly payroll based on basic salary and overtime hours.	
(ii) The programmer writes a pseudocode to plan the step-by-step logic for the user login and authentication process.	
(iii) The programmer intentionally enters alphabet characters into the "Phone Number" text field to check if the program triggers the correct error message.	
(iv) The programmer translates the previously created pseudocode into actual C++ source code using an Integrated Development Environment (IDE).	
(v) The programmer creates a comprehensive user manual and adds internal comments inside the source code for future maintenance references.	

[5 Marks]

c. Read the scenario below and identify the **Input, Process,** and **Output (IPO).**

i. A local cinema wants to create a simple program to calculate the total ticket price for customers. The program needs the customer's age and the number of tickets they want to buy. The standard price for one ticket is RM15. If the customer is below 18 years old, they get a RM3 discount per ticket. The program will display the final total price to be paid.

[3 Marks]

Input	
Process	
Output	

ii. A library system calculates overdue fines for borrowed books. The program requires the type of book (Regular or Reference) and the number of days overdue. The fine rate is RM0.50 per day for Regular books and RM1.00 per day for Reference books. If the book is overdue for more than 7 days, an additional penalty of RM5.00 is added. The system will display the total fine to be paid.

[3 Marks]

Input	
Process	
Output	

2. Developing an algorithm is a crucial process in software development.

   a. What is the definition of an **algorithm**?

[2 Marks]

_____

_____

_____

   b. Based on the textual flowchart description below, write the equivalent **pseudocode**.

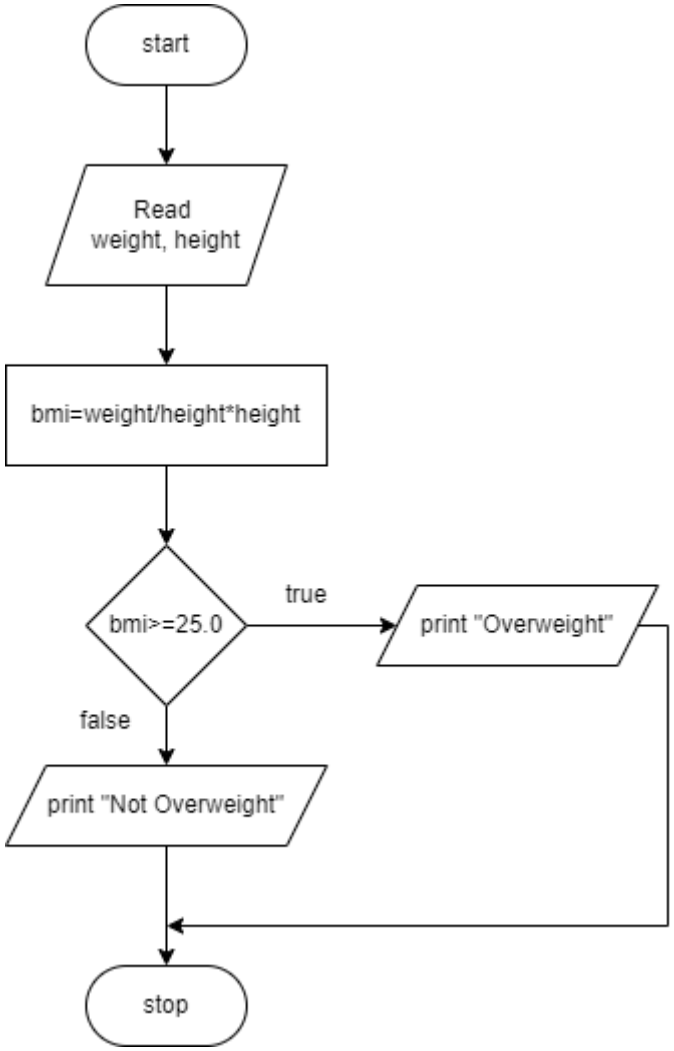


Figure 1

[5 Marks]

3. **Procedural programming** and **object-oriented programming** are examples of programming paradigms.

   a. State one (1) difference between **Procedural Programming** and Object-Oriented **Programming (OOP).**

   [4 Marks]

	Procedural Programming	**Object-oriented Programming**
Difference		

b. Determine whether the following identifiers are **valid** or **invalid** in Python.

[5 Marks]

Identifier	Valid/Invalid
`student_age`	
`1stPrize`	
`total-price`	
`class`	
`student age`	
`MyVar123`	

c. State the appropriate Python data type for the following values:

[3 Marks]

Variable Name	Data Type
`"False"`	
`105.4`	
`True`	

d. Convert the following mathematical equation into a valid Python statement.

[2 Marks]

i. $y=\frac{(a+b)}{c^2+10}$

_____

_____

4.

a. Define single selection control structure.

[1 Mark]

_____

_____

_____

b. Trace the output for the following Python code snippet:

[6 Marks]

```
x = 17
y = 4
z = x % y
w = x // y
print("Value 1 is", z)
print("Value 2 is", w)
```

Figure 2

d. Write a Python if-elif-else segment that assigns a grade based on a student`s mark according to the table below:

Mark	**Grade**
80 – 100	A
60 – 79	B
Bellow 60	C

[4 Marks]

5.

a. Convert the following **for** loop into an equivalent **while** loop.

```
total=0
for count in range(1,10,2):
total=total+count
print("Count:", count)
print("Final Total:", total)
```

[5 Marks]

b. Trace the output for the following code.

[3 Marks]

```
x = 10
while x > 5:
    print(x)
    x = x - 2
```

6. Write a complete Python program for a clinic to process patient`s body temperatures.
   The program should continuously ask the user to enter a patient's body temperature.
   The looping must stop when a sentinel value of 0 is entered.
   Once the loop stops, the program must calculate and display:
   a. The total number of patients recorded.
   b. The average temperature of the patients.

[10 Marks]

END OF QUESTION PAPER